

Information

Presenter

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Work Objectives and Tasks

The objective of this laboratory work is to familiarize with the document structure in LaTeX.

Tasks: Review and work with the document structure in TexLive:

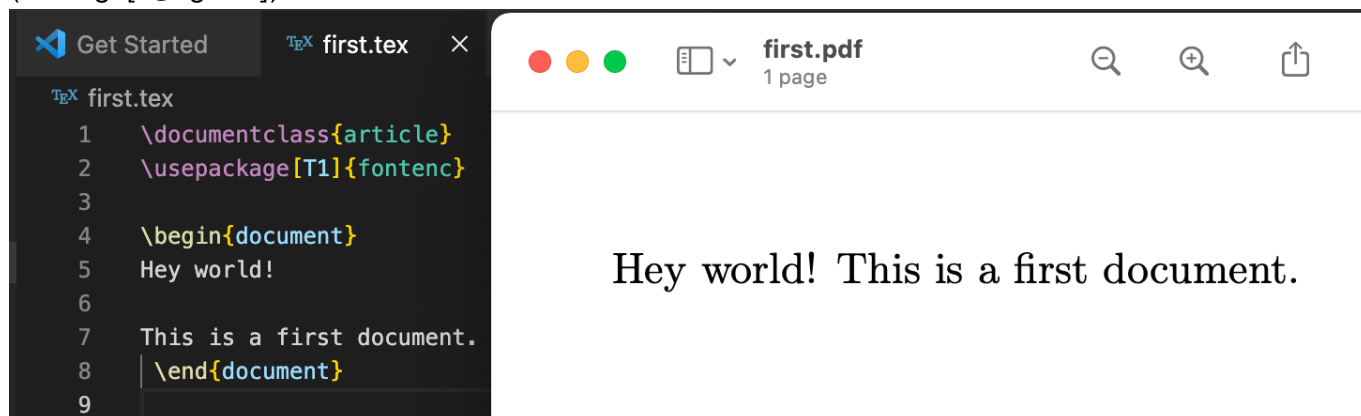
1. Try adding text to your first document, typesetting and seeing the changes in your PDF.
2. Make some different paragraphs and add variable spaces.
3. Explore how your editor works; click on your source and find how to go to the same line in your PDF.
4. Try adding some hard spaces and see how they influence line-breaking.

Theoretical Introduction

Basic LaTeX Document Structure

```
\documentclass{article}
\usepackage[T1]{fontenc}
\begin{document}
Hey world!
This is a first document.
\end{document}
```

(see Fig. [-@fig:001])



{ #fig:001 width=100% }

Running LaTeX

Compilation Process

(see Fig. [-@fig:002])

```
[nadiaezzakate@Nadias-MacBook-Air latex-lab % pdflatex first.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.27 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./first.tex
LaTeX2e <2024-11-01> patch level 2
L3 programming layer <2025-01-18>
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/06/29 v1.4n Standard LaTeX document class
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(/usr/local/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file first.aux.

[1{/usr/local/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}{/usr/l
ocal/texlive/2025/texmf-dist/fonts/enc/dvips/cm-super/cm-super-t1.enc}]
(./first.aux) )</usr/local/texlive/2025/texmf-dist/fonts/type1/public/cm-super/
sfrm1000.pfb>
Output written on first.pdf (1 page, 15651 bytes).
```

{ #fig:002 width=100% }

(see Fig. [-@fig:003])

```
nadiaezzakate@Nadias-MacBook-Air lab02 % pdflatex first.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.27 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./first.tex
LaTeX2e <2024-11-01> patch level 2
L3 programming layer <2025-01-18>
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/06/29 v1.4n Standard LaTeX document class
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/size10.clo))
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(/usr/local/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
No file first.aux.

[1{/usr/local/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}{/usr/l
ocal/texlive/2025/texmf-dist/fonts/enc/dvips/cm-super/cm-super-t1.enc}]
(./first.aux) )</usr/local/texlive/2025/texmf-dist/fonts/type1/public/cm-super/
sfrm1000.pfb>
Output written on first.pdf (1 page, 15644 bytes).
```

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Laboratory Work Execution

Task 1: Creating First Document

Adding Text and Typesetting

(see Fig. [-@fig:004])

{ #fig:004 width=100% }

```
\documentclass[a4paper,12pt]{article}
\usepackage[T1]{fontenc}
```

```
\begin{document}

\title{My First LaTeX Assignment}
\author{Nadia Ezzakate}
\date{\today}

\maketitle

% --- Introduction ---
\section*{Introduction}
Hey world!
This is my first document in \LaTeX.
I want to learn how to structure text, use formatting, and create a clean
PDF.

% --- Paragraphs with formatting ---
\section*{Main Content}
Here is a paragraph with some \textbf{bold text} and some \textit{italic
text}.
It helps emphasize important points.

Another paragraph shows how LaTeX treats      multiple spaces:
extra spaces are automatically reduced to a single space in the output
PDF.

% --- Hard spaces ---
Sometimes we need words to stay together:
Dr.\~Brown lives in New\~York\~City.
The tilde (~) keeps these words on the same line.

% --- Lists ---
Here is a small list of things I want to remember about LaTeX:
\begin{itemize}
    \item Paragraphs are separated by blank lines.
    \item Special characters like \%, \$, \#, \_ must be escaped.
    \item Footnotes add extra information without breaking the
text\footnote{This is an example footnote.}.
    \item Hard spaces (~) prevent unwanted line breaks.
\end{itemize}

% --- Conclusion ---
\section*{Conclusion}
Overall, creating this document helped me understand the basics of LaTeX:
paragraphs, formatting, lists, footnotes, and spacing.
I am ready to explore more advanced features in the next assignments.

\end{document}
```

(see Fig. [-@fig:005])

My First LaTeX Assignment

Nadia Ezzakate

September 27, 2025

Introduction

Hey world! This is my first document in $\text{\LaTeX}$. I want to learn how to structure text, use formatting, and create a clean PDF.

Main Content

Here is a paragraph with some **bold text** and some *italic text*. It helps emphasize important points.

Another paragraph shows how LaTeX treats multiple spaces: extra spaces are automatically reduced to a single space in the output PDF.

Sometimes we need words to stay together: Dr. Brown lives in New York City. The tilde () keeps these words on the same line.

Here is a small list of things I want to remember about LaTeX:

- Paragraphs are separated by blank lines.
- Special characters like %, \$, #, _ must be escaped.
- Footnotes add extra information without breaking the text¹.
- Hard spaces () prevent unwanted line breaks.

Conclusion

Overall, creating this document helped me understand the basics of LaTeX: paragraphs, formatting, lists, footnotes, and spacing. I am ready to explore more advanced features in the next assignments.

¹This is an example footnote.

Creating Different Paragraph Structures

(see Fig. [-@fig:006])

Main Content

Here is a paragraph with some **bold text** and some *italic text*. It helps emphasize important points.

Another paragraph shows how LaTeX treats multiple spaces: extra spaces are automatically reduced to a single space in the output PDF.

Sometimes we need words to stay together: Dr. Brown lives in New York City. The tilde () keeps these words on the same line.

Here is a small list of things I want to remember about LaTeX:

- Paragraphs are separated by blank lines.
- Special characters like %, \$, #, _ must be escaped.
- Footnotes add extra information without breaking the text¹.
- Hard spaces () prevent unwanted line breaks.

{ #fig:006 width=100% }

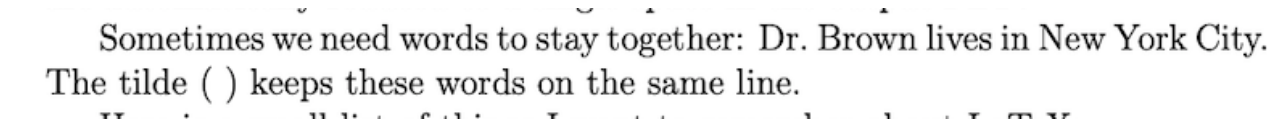
Task 3 & 4: Editor Exploration and Hard Spaces

Editor Features and Line-breaking Control

```
26  % --- Hard spaces ---
27  Sometimes we need words to stay together:
28  Dr.~Brown lives in New~York~City.
29  The tilde (~) keeps these words on the same line.
```

{ #fig:007 width=100% }

(see Fig. [-@fig:008])



{ #fig:008 width=100% }

Conclusion

Thus, the objective of installing TeXlive was achieved, and I became familiar with working with its structure as well as with the LaTeX system. The laboratory work allowed me to practice creating LaTeX documents, understanding paragraph formatting, and using non-breaking spaces to control line-breaking behavior.

{.standout}

Thank you for your attention